Computer Networks: OSI 7-Layer vs TCP/IP Architecture

Protocol stack, IP addressing, subnetting and routing algorithms

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1. Overview of OSI Model: Physical, Data Link, Network, Transport, Session, Presentation, Application.
2. TCP/IP Model and Comparison with OSI.
3. Data Link Layer: Framing, Error Detection CRC, Flow Control Sliding Window.
4. Network Layer: IPv4 vs IPv6 header format, CIDR notation and Subnetting.
5. Routing Protocols: Distance Vector RIP vs Link State OSPF.
6. Transport Layer: TCP 3-Way Handshake, Flow Control, Congestion Control vs UDP.
7. Application Layer Protocols: HTTP, HTTPS, DNS, DHCP, FTP.